习题16.1:第5,6, 11
习题16.2:第9,10，又:试给出重积分的“第二中值定理”
习题16.3:第1 (2,5),2 (1,5),5, 13, 15 (2)
习题16.4:第4,5，10-18中任选四道题.

16.1.5

Consider $f:[0,1]^2\to\mathbb{R},(x,y)\mapsto\begin{cases}\sin(1/xy), & xy>0\\ 0, & xy=0\end{cases}$. Determine whether f is integrable.
Proof: f is bounded, and the discontinuous points belong to the null set $\{(x,y):xy=0\}$, hence form a null set. Therefore f is integrable.

16.1.6

Suppose $f:I=[a,b]\times[c,d]\to\mathbb{R}$ and $f(x,-),f(-,y)$ are monotone increasing for each x,y, prove that $f\in R(I)$.
Proof: Let $A_{n,k}=\left\{x:\sup_{y,z\in B(x,1/k)}|f(y)-f(z)|>\frac{1}{n}\right\}$ and likewise define $B_{n,k}$ and $C_{n,k}$ by replacing f with $f(x,-)$ and $f(-,y)$. By monotone condition, $A_{n,k}\subset B_{n,2k}\cup C_{n,2k}$. Let $M=\sup_I|f|$, for a fixed x, $f(x,-)$ is increasing (hence Riemann integrable) so (if we let $f(x,y)=f(x,d)$ for $y>d$ and $f(x,y)=f(x,c)$ for $y<c$) $0\leqslant\int_c^d f\left(x,y+\frac{1}{n}\right)-f\left(x,y-\frac{1}{n}\right)\,\mathrm{d}y\leqslant\frac{4M}{n}$. Therefore $m(B_{n,k})\leqslant\frac{Cn}{k}$ and analogously $m(C_{n,k})\leqslant\frac{Cn}{k}$ for some constant C.
Therefore $m(A_{n,k})\leqslant m(B_{n,2k})+m(C_{n,2k})\leqslant\frac{Cn}{k}$, so the discontinuous points $\bigcup_n\bigcap_k A_{n,k}$ form a null set, hence $f\in R(I)$.

16.1.11

Let $I=[a,b]\times[c,d]$ and $D=\chi_{\mathbb{Q}^2}\cap I$, prove that D is not integrable.
Proof: D is nowhere continuous, hence not Riemann integrable.

16.2.9

Suppose Ω is open and Jordan measurable, $f\in C(\Omega)$. If for any Jordan set $A\subset\Omega$, $\int_A f\,\mathrm{d}x=0$, show that $f\equiv 0$.
Proof: If $f(x)\neq 0$ suppose $f(x)>0$ and take $B(x,r)\subset\Omega$ such that $f(y)>\frac{1}{2}f(x)\forall y\in B(x,r)$. Then $\int_{B(x,r)}f\,\mathrm{d}x>0$ a contradiction, so $f\equiv 0$.

16.2.10

Suppose Ω is open and Jordan measurable, $f\in C(\overline{\Omega})$ and for any $\phi\in C(\overline{\Omega})$ such that $\phi|_{\partial\Omega}=0$, $\int_\Omega f\phi\,\mathrm{d}x=0$. Prove that $f\equiv 0$.
Proof: If $f(x)\neq 0$ for $x\in\Omega$, suppose $f(x)>0$, then take an open ball $B(x,2r)\subset\Omega$ such that $f(y)>\frac{1}{2}f(x)\forall y\in B(x,2r)$. Take $\phi\in C_c^\infty(\overline{\Omega})$ such that$0\leqslant\phi\leqslant 1$, $\mathrm{supp}\phi\subset\overline{B}(x,2r)$ and $\phi|_{B(x,r)}\equiv 1$. Then $\phi|_{\partial\Omega}=0$ and $\int_\Omega f\phi\,\mathrm{d}x>\int_{B(x,r)}f\,\mathrm{d}x>0$ a contradiction. Hence $f|_\Omega=0$, so by continuity $f\equiv 0$.

16.3.1

(2) $I=[0,\pi/2]^2$,

$$\begin{aligned}\int_I \sin(x+y)\,\mathrm{d}x\mathrm{d}y &= \int_0^{\pi} \sin t\cdot\left(\frac{\pi}{2}-\left|\frac{\pi}{2}-t\right|\right)\,\mathrm{d}t = 2\int_0^{\pi/2} t\cos t\,\mathrm{d}t\\ &= \pi - 2\int_0^{\pi/2}\sin t\,\mathrm{d}t = \pi-2.\end{aligned}$$

(5) $A=\left\{(x,y): y^2\leqslant 2px, x\leqslant \frac{p}{2}\right\}$,

$$\begin{aligned}\int_A xy^2\,\mathrm{d}x\mathrm{d}y &= \int_0^{p/2} x\int_{-\sqrt{2px}}^{\sqrt{2px}} y^2\,\mathrm{d}y\,\mathrm{d}x = \int_0^{p/2}\frac{(2p)^{3/2}}{3}x^{5/2}\,\mathrm{d}x\\ &= (2p)^{3/2}\frac{2}{21}(p/2)^{7/2} = \frac{p^5}{42}.\end{aligned}$$

16.3.2

(1) $\Omega=\left\{(x,y,z): \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\leqslant 1\right\}$.

$$\int_\Omega x+y+z\,\mathrm{d}x\mathrm{d}y\mathrm{d}z = abc\int_{D^3} ax+by+cz\,\mathrm{d}x\mathrm{d}y\mathrm{d}z = abc(a+b+c)\int_{D^3} x\,\mathrm{d}x\mathrm{d}y\mathrm{d}z = 0.$$

(5) $\Omega=\left\{(x,y,z): 0\leqslant z\leqslant 64, x^2+y^2\in\left[\frac{z}{16},\frac{z}{4}\right]\right\}$.

$$\begin{aligned}\int_\Omega x^2+y^2\,\mathrm{d}x\mathrm{d}y\mathrm{d}z &= \int_0^{64}\int_{\sqrt{z}/4}^{\sqrt{z}/2} r^2\cdot 2\pi r\,\mathrm{d}r\,\mathrm{d}z = 2\pi\int_0^{64}\frac{3z^2}{64}\,\mathrm{d}z\\ &= \frac{3\pi}{32}\frac{1}{3}64^3 = 2^{13}\pi.\end{aligned}$$

16.3.5

Calculate the volume of $\Omega=\{|x|+|y|+|z|\leqslant a\}$.
Solution: Clearly $m(\Omega)=8a^3m(\{(x,y,z): x+y+z\in[0,1]\})$, and the volume of the 3-simplex is $1/3!$, so $m(\Omega)=\frac{4a^3}{3}$.
(The volume of the n-simplex is $1/n!$, since the unit cube decomposes into $n!$ parts $R_\sigma=\{0<x_{\sigma(1)}<\cdots<x_{\sigma(n)}<1\}$), and $\{0<x_1<\cdots<x_n<1\}$ can be mapped to the n-simplex via the transformation $y_1=x_1, y_i=x_i-x_{i-1}$ with determinant one.

16.3.13

Suppose $f\in R[a,b]$, prove that

$$\int_a^b f(x_1)\,\mathrm{d}x_1\int_a^{x_2} f(x_2)\,\mathrm{d}x_2\cdots\int_a^{x_{n-1}} f(x_n)\,\mathrm{d}x_n = \frac{1}{n!}\left(\int_a^b f\,\mathrm{d}x\right)^n.$$

Proof: See the previous exercise, we showed that $[a,b]^n$ can be decomposed into $n!$ isomorphic pieces of n-simplexes.

16.3.15(2)

Calculate $\int_{[0,1]^n} \max\{x_i\}\,\mathrm{d}x_1\mathrm{d}x_2\cdot\mathrm{d}x_n$.
Solution: By the layer-cake formula, let $M(x)=\max\{x_i\}$, then

$$\int_I M(x)\,\mathrm{d}x=\int_0^1 m(\{x:M(x)>t\})\,\mathrm{d}t=\int_0^1 1-t^n\,\mathrm{d}t=\frac{n}{n+1}.$$

16.4.4&5

Calculate the volume of $A=\left\{(x_i)\in\mathbb{R}^n:\sum_i\left(\sum_j a_{ij}x_j\right)^2\leqslant h^2\right\}$ where $\det(a_{ij})\neq 0$.
Solution: Let $T=(a_{ij})$, then $A=\{(x_i)\in\mathbb{R}^n:\|Tx\|^2\leqslant h^2\}$ so $A=T^{-1}(B(0,h))$. By polar decomposition, each invertible T is the product of an orthogonal matrix and a diagonal one, hence the volume of A is just $m(B(0,h))/|\det T|$, which is $\alpha(n)h^n/|\det(a_{ij})|$.

16.4.15

Suppose $f\in C(\mathbb{R})$, prove that $\int_{|x|+|y|\leqslant 1} f(x+y)\,\mathrm{d}x\mathrm{d}y=\int_{-1}^1 f(u)\,\mathrm{d}u$.
Proof: Let $u=x+y, v=x-y$, then $|x|+|y|\leqslant 1$ is just $\max\{|u|,|v|\}\leqslant 1$. Hence $\int_{|x|+|y|\leqslant 1} f(x+y)\,\mathrm{d}x\mathrm{d}y$ $=\int_{-1}^1 f(u)\,\mathrm{d}u$.

16.4.16

Prove that $\int_{x^2+y^2\leqslant 1} f(ax+by)\,\mathrm{d}x\mathrm{d}y=2\int_{-1}^1\sqrt{1-u^2}f(ku)\,\mathrm{d}u$ where $k=\sqrt{a^2+b^2}$.
Proof: Let $u=ax+by, v=-bx+ay$ then

$$\int_{x^2+y^2\leqslant 1} f(ax+by)\,\mathrm{d}x\mathrm{d}y=k^{-2}\int_{u^2+v^2\leqslant k^2} f(u)\,\mathrm{d}u\mathrm{d}v=2\int_{-1}^1\sqrt{1-u^2}f(ku)\,\mathrm{d}u.$$

16.4.17

Prove that $\int_{x^2+y^2+z^2} f(ax+by+cz)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z=\pi\int_{-1}^1(1-u^2)f(ku)\,\mathrm{d}u$, where $k=\sqrt{a^2+b^2+c^2}$.
Proof: Analogously, let $u=\frac{1}{k}(ax+by+cz)$, and take orthonormal basis u,v,w such that $dxdydz$ $=dudvdw$ and $x^2+y^2+z^2=u^2+v^2+w^2$. Hence change of variables gives the desired formula .

16.4.18

Calculate the Jacobian of the polar transformation in $\mathbb{R}^n$.
Solution: Recall the transformation is $\mathbb{R}^n\to(0,\infty)\times[0,\pi]^{n-2}\times[0,2\pi)$, given by

$$\begin{cases} x_1 = r\cos\theta_1, \\ x_2 = r\sin\theta_1\cos\theta_2, \\ \cdots \\ x_{n-1} = r\sin\theta_1\cdots\sin\theta_{n-2}\cos\theta_{n-1}, \\ x_n = r\sin\theta_1\cdots\sin\theta_{n-1}. \end{cases}$$

So the Jacobian is $r^{n-1}\sin^{n-2}\theta_1\cdots\sin\theta_{n-2}$.